Data Science for Biologists: A Hands-On Introduction with R and Bioconductor

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Abstract

Background. Biomedical researchers are increasingly expected to analyze their own data reproducibly, yet most have had no formal training in statistics or data science, and generic "intro to R" materials teach syntax without the conceptual grounding that biological data analysis requires.

Methods. *The RBioc Book* is an open-access, fully executable textbook, built as a Quarto book on the `knitr` engine so that every chapter runs its own R code at render time and prose, code, and output are always in sync. It is designed around adult-learning principles, carries biological framing from the first chapters, and builds a cumulative conceptual scaffold including R programming, data structures, and statistical reasoning that the genomics and single-cell chapters later reuse. It is openly licensed (CC-BY 4.0), versioned, and archived with a citable DOI.

Results. The book takes a reader from a first line of R to the analysis of real genomic data with Bioconductor, teaching against public datasets. It interleaves conceptual explanations, statistical and data science concepts, and biological context. It has been taught in four formal course offerings (~100 trainees, 2025–2026) and, as an online resource, recorded 7,513 page views between May 2024 and June 2026 — with 876 unique readers reaching the landing page alone — and per-chapter active-engagement times indicating readers work through rather than skim the material. Instructors can adopt a chapter as a lab, fork it for a local course, or assign it for self-paced study; every result is reproducible by re-running the code.

Conclusions. *The RBioc Book* is a feature-complete, reusable, openly-licensed teaching resource that closes the concept-and-reproducibility gap for biologists endeavoring to understand data science, statistics, and reproducible computational research. It is also a template for large-scale executable, community-maintained teaching materials in the Bioconductor ecosystem.

Introduction

Biologists increasingly need to understand statistical and data science principles, but many biologists have limited formal coursework in statistics or data science (Drnevich et al. 2025). At the same time they are expected to drive analyses of their own data — and to do so reproducibly. Many generic "intro to R" courses show *how* to perform an analysis but do not motivate it with concept and rationale, leaving learners able to copy a workflow but not to reason about it or adapt it to their own problems.

The RBioc Book is an open-access, executable textbook that takes a reader from their first line of R (R Core Team 2024) to the analysis of real genomic data with Bioconductor (Huber et al. 2015). It is written for biomedical researchers who know the biology and have been introduced to programming, statistics, and computation but want a more formal and thorough treatment of the concepts and skills; a biology primer makes it usable from the other direction as well, for computational readers approaching biological data for the first time. Rather than teaching programming in the abstract and leaving the biology for "later," the book carries biological framing from the first chapters and builds the conceptual scaffolding that the genomics chapters later depend on. Because the book is fully executable and openly licensed, an instructor can adopt a chapter as a lab, fork it for a local course, or assign it for self-paced study, and a learner can reproduce every result by running the same code.

Implementation

The book is built as a Quarto book (Allaire et al. 2024) so that rendering *executes* the R code in each chapter: the prose, the code, and the output the reader sees are produced from the same source and are always in sync. This executable design is at the core of the book. It makes the entire book a reproducible artifact rather than a static document, and it means that keeping the book correct is the same act as keeping the code runnable.

The material is organized around adult-learning principles (Knowles, Holton III, and Swanson 2015). Each chapter opens with explicit learning objectives, motivates concepts before formalizing them, and pairs worked examples with exercises and solutions. A consistent callout taxonomy (objectives, notes, warnings, exercises) gives the reader predictable signposts throughout. The narrative is intentionally cumulative: early chapters earn the vocabulary and data structures that the statistics, machine-learning, and genomics chapters reuse. The genomics material is taught against real, public datasets and the Bioconductor classes practitioners actually use, so the skills transfer directly to the reader's own analyses.

Operation

Minimal requirements. Reading the book online requires only a web browser (https://seandavi.github.io/RBiocBook). Building or extending it locally requires R with the packages declared in the repository's `DESCRIPTION`, and Quarto; the `knitr` engine executes each chapter's R at render time.

Workflow. From a clone of the repository:

```
quarto render                        # all formats (html, pdf, epub) into _book/
quarto preview                       # live-reload server while editing
quarto render <file>.qmd --to html   # single chapter, fast iteration
```

Adoption modes. Because each chapter is a self-contained, executable `.qmd` document, an instructor can (i) assign the online edition for self-paced study, (ii) adopt an individual chapter as a lab, or (iii) fork the repository and adapt chapters for a local course, re-rendering to regenerate every result. The book is versioned and each release is archived with a citable DOI, so a course can pin to a specific version.

Use cases

The RBioc Book is designed to support both synchronous and asynchronous teaching: learners ingest material at their own pace, freeing instructors to engage participants directly rather than lecture syntax.

Executable chapter (input → output). A representative chapter loads a public dataset, constructs the relevant Bioconductor container (e.g., a `SummarizedExperiment`), and produces the summary or figure inline; the reader reproduces the output by rendering the same chapter, with no hidden state. This input (chapter source + declared R environment) → output (rendered prose, code, and result) round-trip is the unit the book is built from.

Formal course use. The book has been taught in four formal offerings — the Cold Spring Harbor Laboratory data-analysis course (2025 and 2026) and the BigCARE program (2025 and 2026), each with roughly 25 participants per session (about 100 trainees over two years).

Open readership. Beyond the classroom, between May 2024 and June 2026 the online edition recorded 7,513 page views across chapters, with 876 unique readers reaching the landing page alone, and substantial per-chapter engagement (for example, the single-cell setup chapter averaged over seven minutes of active engagement per reader), indicating that readers work through the material rather than skim it. The book continues to be developed in the open and contains social coding best practice guidance to accept contributions.

Summary

The RBioc Book closes the gap between "how to run R" and "how to reason about, and reproducibly perform, biological data analysis," in one continuous path from a first line of R to real Bioconductor genomics. Its executable, openly-licensed, version-archived design makes it a reusable teaching resource — adoptable whole or in part — and a template for community-maintained, reproducible teaching materials in the Bioconductor ecosystem.

Data availability

The book teaches against publicly available datasets, cited and linked in place within the relevant chapters. Website usage figures reported here were derived from Google Analytics 4 for the book site over May 2024–June 2026; the underlying per-page export used for the reported totals is included in the source repository (`paper/data/`).

Software and resource availability

- **Source code / book repository:** https://github.com/seandavi/RBiocBook
- **Live book:** https://seandavi.github.io/RBiocBook
- **Archived version (concept DOI):** https://doi.org/10.5281/zenodo.20574829
- **License:** Original content (prose, original figures, example code) is released under CC-BY 4.0; example code is additionally reusable as such. Third-party figures, datasets, and adapted passages retain their own open licenses, each documented on the book's *Contributors & acknowledgments* page.
- **Machine-readable citation:** `CITATION.cff` in the repository.

Author contributions

Lori Shepherd Kern — authored *Base R versus the tidyverse*, *Control statements*, *Factors*, and *Organizing, saving, and loading your work*, with edits throughout. Sean Davis — conceived the book; primary author and editor; wrote most chapters. Both authors reviewed and approved the manuscript.

Competing interests

No competing interests were disclosed.

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